DAY 1 LEARNINGS-TASK1

WHAT IS TIME SERIES ANALYSIS?

Example of a time series graph:
No. of hours you sleep vs days

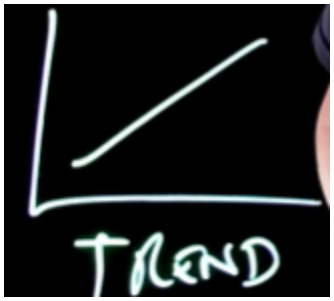
Data for the same entity, which is the hours a person has slept, is collected at regular intervals, like over days.

We analyze the timestamp data to extract the meaningful insights and predictions about the future.

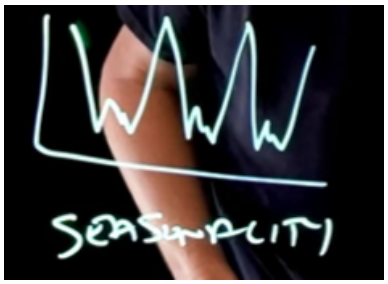
Helps organizations to make business decisions. **For example**, a retailer can use this functionality to predict the future sales and optimize the inventory levels.
Another one- A purchaser can use this to predict the commodity prices and make informed purchasing decisions.
In agriculture, predict weather patterns, influencing decisions on harvesting and when to plant.

COMPONENTS

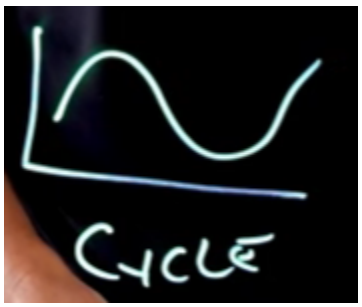
1. **TREND:**
   Overall direction of the data over time, whether it's increasing/decreasing or perhaps the same



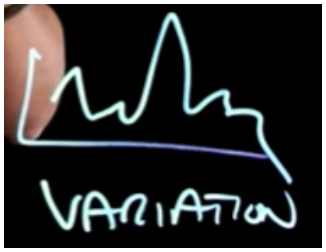
2. **SEASONALITY**:
   Repeating the pattern of the data set over a period of time, retail sales increase during the holiday season. Spikes observed.



3. **CYCLE:**
   Repeating but non seasonal patterns in the data. Economic booms or busts that happen over several years/decades



4. **VARIATION:**
   These other components cannot explain the unpredictable ups and downs in the data. Irregularities/Noise.



FORECASTING MODELS

1. **ARIMA model**
   Auto Regressive Integrated Moving Average
   AR- How past values affect future values
   I- Differencing component- trends & seasonalities
   MA- Smooths the noise by removing the non deterministic or random movements from time series
2. **EXPONENTIAL SMOOTHING**
   Forecast the time series data that does not have a clear trend or seasonality.
   Smoothing out the area, more weight to the recent values and less weight to the older values

Various models out there
Choosing one depends on:

1. **Data**
2. **Specific problem**

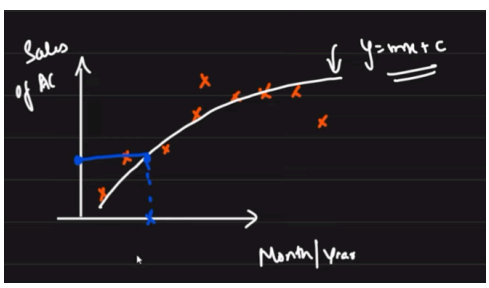
IMPLEMENTATION

1. Python-pandas (statistical analysis), matplotlib(visualization)
2. R
3. MATLAB

Time Series Vs Non Time Series Problems- Why Time Series Forecasting Is Difficult? Why is TSA important?

Time Series Data—Data points collected/recorded at successive points in time, in regular intervals (e.g., stock prices, weather, sales data).

Non-Time Series data has no temporal ordering — observations are independent (e.g., predicting house prices based on features).



Interpolation problem statements

Key Differences

Aspect	Time Series	Non-Time Series
Data order	Matters (sequential)	Doesn't matter

Observations	Dependent on past values	Independent
Split method	Must split chronologically	Random split is fine
Main goal	Forecasting future	Classification/Regression

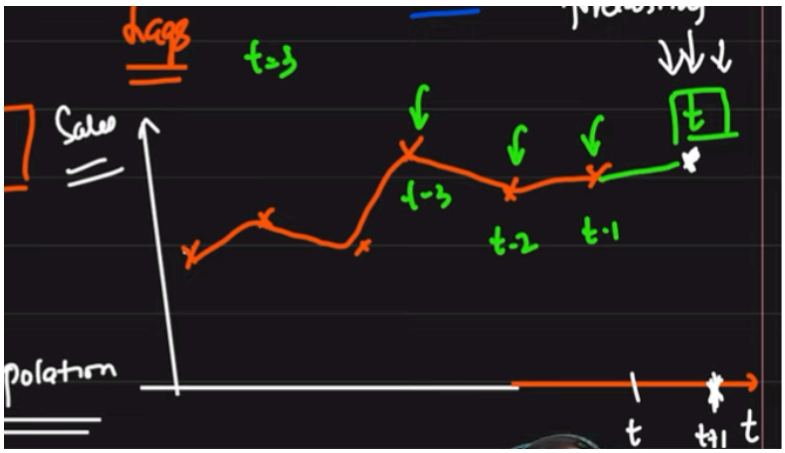
What makes Time Series Forecasting challenging?

- **Temporal Dependency** — Past values influence future values; you can't treat data as independent.
- **Trend** — Data may rise/fall over time, making patterns shift.
- **Seasonality** — Repeating patterns at fixed intervals (daily, weekly, yearly) add complexity.
- **Irregularity/Noise** — Random fluctuations that are hard to model.
- **Non-Stationarity** — Mean, variance, and covariance change over time; most models require stationarity.
- **Multiple Seasonalities** — E.g., electricity demand has hourly, daily, and weekly patterns simultaneously.
- **Autocorrelation** — Observations correlate with their own past values, violating standard ML assumptions.
- **External Shocks** — Unexpected events (COVID, recessions) can break historical patterns entirely.
- **Limited Data** — Collecting long historical sequences takes real time; can't just gather more samples.
- **Overfitting Risk** — Models can memorize past patterns that won't repeat.

Why is Time Series Analysis (TSA) Important?

- **Business Forecasting** — Sales, demand, revenue prediction for planning and inventory.
- **Finance** — Stock price trends, risk modeling, algorithmic trading.
- **Healthcare** — Patient monitoring, disease outbreak prediction (e.g., flu seasons).
- **Economics** — GDP, inflation, unemployment trend analysis.
- **Weather Forecasting** — Critical for agriculture, disaster management.
- **Operations** — Server load prediction, supply chain optimization.

- **Anomaly Detection** — Spot unusual spikes (fraud, system failures) in real time.
- **Policy Making** — Governments use economic time series to shape decisions.



INTERPOLATION VS. EXTRAPOLATION????